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High-Accuracy Performance of a Simplified Binary Triage System in Military Trauma

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IMPORTANCE Accurate prehospital triage is critical for prioritizing evacuation and allocating medical resources in combat settings characterized by diagnostic uncertainty and operational constraints. Simplified, judgment-based triage systems may improve feasibility and adherence but require validation under real-world conditions.

OBJECTIVE To evaluate the performance of a simplified binary field triage system in identifying severely injured trauma casualties under real-world combat conditions.

DESIGN, SETTING, AND PARTICIPANTS This cohort study was conducted between October 27, 2023, and January 19, 2025. The setting was a military prehospital medical system during a recent conflict. Included were military trauma casualties treated and evacuated by the Israel Defense Forces Medical Corps.

MAIN OUTCOMES AND MEASURES Field triage designations (urgent vs nonurgent) were compared with anatomic injury severity defined by the Abbreviated Injury Scale (AIS) and Injury Severity Score (ISS), as well as the need for trauma intervention (NFTI). Primary outcomes included undertriage and overtriage rates.

RESULTS Among 4522 casualties (median [IQR] age, 23 [20-29] years; 4479 male [99%]), a judgment-based binary approach was associated with an *undertriage rate*, classification as nonurgent despite severe injury, of 8.5% (95% CI, 6.81%-10.54%) among casualties with any injury with an AIS score of 3 or greater. Among casualties with major trauma (ISS ≥16), the undertriage rate was 3.6% (95% CI, 2.25%-5.67%). Corresponding sensitivities were 91.5% (95% CI, 89.46%-93.19%) and 96.4% (95% CI, 94.33%-97.75%), respectively. *Overtriage*, defined as urgent triage in the absence of severe injury, was 62.7% (95% CI, 60.65%-64.77%). Using the NFTI definition, undertriage was 8.8% (95% CI, 7.33%-10.62%), and overtriage was 50.2% (95% CI, 48.06%-52.32%). In undertriaged patients with major trauma, severe head and thoracic injuries were most common.

CONCLUSIONS AND RELEVANCE In this cohort study, a simplified, judgment-based binary field triage system demonstrated high sensitivity with low undertriage for major trauma in combat settings, at the cost of substantial overtriage. These findings support the use of simple triage approaches under operational constraints while highlighting the need to improve detection of less apparent injuries without compromising simplicity.

Supplemental content

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Timely and accurate triage of trauma casualties is a cornerstone of effective prehospital care.[1] Rapid and accurate identification of injury severity at the scene enables optimal allocation of life-saving interventions and appropriate prioritization for evacuation.[2]

Most existing field triage protocols have been developed within civilian trauma systems and rely on the integration of multiple clinical variables, including physiologic parameters, anatomic injury patterns, and mechanism of injury.[3] Despite their widespread use, these approaches have demonstrated inconsistent performance and wide variability in diagnostic accuracy.[4] Importantly, reliance on numerous data points may limit the feasibility and adherence among medical practitioners in the prehospital settings.[4] These limitations highlight the need for simplified triage approaches.[5]

In combat settings, these challenges are further amplified.[6] Military prehospital care is distinct both in its operational context and in the nature of injury. Triage decisions must frequently be made in austere environments, under ongoing threat, with limited diagnostic capability and constrained evacuation resources.[6] Within this context, simplified triage frameworks may balance the need for rapid decision-making with the uncertainty inherent to combat casualty care.

The Israel Defense Forces Medical Corps (IDF-MC) uses a binary triage system[7] that categorizes casualties as either urgent, indicating risk to life, limb, or critical function, or nonurgent, in the absence of such risk. This approach allows a clear transfer of information between medical and nonmedical personnel. Once classified as urgent, the casualty is prioritized for evacuation and treatment resources, regardless of a specific injury type or mechanism.

The clinical accuracy of this system can be evaluated by comparing prehospital triage designations to injury severity scores assessed retrospectively.[5,8] Two key metrics in this context are overtriage and undertriage. High undertriage rates may be associated with delayed definitive care and increased mortality, whereas high overtriage rates may risk and overburden limited medical and logistical assets.

In this study, we evaluated the performance of the IDF-MC binary field triage system in identifying severely injured casualties under real-world combat conditions.

Key Points

Question How accurately does a simplified judgment-based binary field triage system identify severely injured trauma casualties during combat operations?

Findings In this cohort study of 4522 military trauma casualties, the triage system showed high sensitivity, with low undertriage for major trauma in combat settings at the cost of increased overtriage.

Meaning Findings reveal that the binary triage system showed high sensitivity, supporting simple, judgment-based triage approaches in combat settings.

Methods

Study Design and Data Sources

The study was approved by the IDF-MC institutional review board, which granted a waiver of informed consent due to the retrospective records-based nature of the study. This retrospective cohort study evaluated the accuracy of the simplified IDF-MC field triage system. Triage designations were compared with the Injury Severity Score (ISS).[9] *Severe injury* was defined as the presence of at least 1 Abbreviated Injury Scale (AIS) score of 3 or greater. Special focus was placed on casualties with ISS equal to or greater than 16, defined as a threshold for major trauma.[10] To assess robustness across alternative definitions of injury severity, triage designations were additionally compared with the need for trauma intervention (NFTI) definition.[5,11,12] This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines.

Demographic, clinical, and operational data were extracted from the Israel Defense Forces Trauma Registry (IDF-TR), which systematically captures all trauma cases managed by IDF medical personnel.[13] Data are sourced primarily from casualty cards completed at point of injury. The forms may be completed in paper format or via digital platforms that replicate the same standardized structure, as previously described.[7,14] Additionally, prehospital health care practitioners are debriefed by trained personnel, typically within 24 hours. Information from these debriefings is used to complete missing structured fields and resolve ambiguities in documentation. Urgency classification is recorded in a dedicated field within the registry.

In-hospital data, including ISS and trauma interventions, were obtained from the Israeli National Trauma Registry (INTR).[15] The INTR captures admissions to all designated trauma centers across Israel, including all facilities receiving military casualties evacuated from the field. It includes patients hospitalized within 72 hours of sustaining physical trauma, excluding poisoning, drowning, or suffocation. Patients who died before hospital arrival or were discharged following emergency department (ED) evaluation without hospitalization are not included in the INTR. Consequently, ISS scores were unavailable for patients discharged directly from the ED. Participant race and ethnicity data were not routinely collected in the study data sources and were therefore unavailable for analysis.

IDF Field Triage

The field triage system is a simplified binary classification in which casualties are designated as urgent when there is a suspected risk to life, limb, or critical function, and as nonurgent otherwise.[7] Its primary purpose is to provide a clear communication framework between medical and nonmedical personnel.

In operational settings, triage decisions are made by advanced life support (ALS) practitioners (paramedics or physicians) embedded within frontline units. These practitioners undergo standardized training, ensuring a consistent clinical approach across practitioner levels.[16]

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Triage is guided by clinical judgment within an operational framework that integrates medical and tactical priorities, beginning with a rapid primary assessment and refined through ongoing evaluation. An example of triage decision-making includes classifying high-velocity penetrating trauma as urgent due to risk of occult life-threatening injury, except for clearly isolated distal injuries without apparent neurovascular compromise. Other examples include assigning urgent status in the presence of high-risk features such as impaired consciousness, hemodynamic instability, or respiratory compromise. Designation may be modified at any stage. Among casualties designated as urgent, evacuation priority is determined by relative injury severity and operational considerations, particularly in mass-casualty events.

Definitions of Overtriage and Undertriage

Overtriage was defined as the proportion of patients without severe injury of all patients designated as urgent (1 – positive predictive value).[8,17,18] Patients with nonsevere injuries included those discharged from the ED without hospitalization, who were considered unlikely to have sustained severe injuries despite the absence of ISS data, as well as hospitalized patients with minor injuries (absence of any AIS score of 3 or greater). To further characterize overtriage and operational prioritization, additional analyses were performed among hospitalized patients classified as urgent to examine the association between injury severity and evacuation time.

Undertriage was defined as the proportion of patients classified as nonurgent from all patients with severe injury (1 – sensitivity).[8,17,18] Patients discharged from the ED without hospitalization were assumed unlikely to have sustained severe injuries and were therefore excluded from the undertriage analysis. In addition, we examined operational and clinical factors associated with undertriage, with a specific focus on patients with major trauma (ISS ≥16).

In analyses using the NFTI definition, casualties were classified as having severe injury if they met at least 1 of the following criteria: early blood product transfusion in the field or ED, operative intervention within 90 minutes, interventional radiologic procedures or intensive care unit admission from the ED, mechanical ventilation in the ED, or death within 60 hours.[5,11,12]

Statistical Analysis

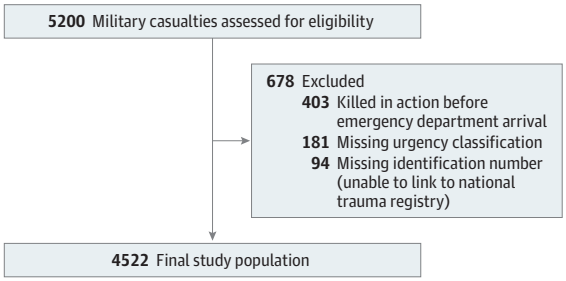
Categorical variables were presented as counts and percentages and compared using the χ^2 test or Fisher exact test, as appropriate. Continuous variables were expressed as medians with IQRs and compared using the Mann-Whitney *U* test. Undertriage and overtriage rates were presented with 95% CIs. A 2-sided *P* value < .05 was considered statistically significant. All analyses were conducted using R, version 4.3.1 (R Core Team).

Results

Patient Characteristics by Field Urgency Classification

Of 5200 military casualties treated and evacuated by the IDF-MC during the study period, excluded were 403 casualties who were killed in action before arrival at the ED, 181 with unknown field urgency classification, and 94 for whom a personal identification number was unavailable (precluding linkage with the INTR). A total of 4522 military casualties were included (**Figure 1**). The cohort was predominantly male (4479 [99%]), with 43 female (1.0%) casualties, and a median (IQR) age of 23 (20-29) years (**Table 1**). The number of casualties per event differed significantly between urgent and nonurgent casualties (*P* < .001), eg, with urgent casualties involved in events with 4 or more casualties in 1093 of 2112 (51.8%) vs 703 of 2410 (29.2%). Similarly, the number of injured body regions identified in the field differed significantly between groups (*P* < .001), eg, with urgent casualties sustaining injuries involving 3 or more regions in 484 of 2112 casualties (22.9%) vs 136 of 2410 nonurgent casualties (5.6%). Injury mechanisms also differed significantly (*P* < .001), eg, with urgent casualties injured by explosive mechanisms in 1527 of 2095 (72.9%) vs 1370 of 2376 (57.7%), by firearms in 413 of 2095 (19.7%) vs 173 of 2376 (7.3%), and by blunt mechanisms or falls in 87 of 2095 (4.2%) vs 598 of 2376 (25.2%). Injuries to the head (300 of 2112 [14.2%] vs 219 of 2410 [9.1%]), neck (272 of 2112 [12.9%] vs 106 of 2410 [4.4%]), torso (672 of 2112 [31.8%] vs 277 of 2410 [11.5%]), pelvis (247 of 2112 [11.7%] vs 93 of 2410 [3.9%]), and lower extremities (895 of 2112 [42.4%] vs 728 of 2410 [30.2%]) were more frequently identified in the field among urgent casualties (all *P* < .001).

Figure 1. Flow Diagram Depicting the Selection of the Study Population



Flow diagram of military trauma casualties treated by the Israel Defense Forces Medical Corps during the study period. Exclusions included death before emergency department arrival, missing field urgency classification, or inability to link to the national trauma registry.

Prehospital interventions, including tourniquet application (593 of 2112 [28.1%] vs 54 of 2410 [2.2%]; *P* < .001), advanced airway management (eg, endotracheal intubation, 66 of 2112 [3.1%] vs 0; *P* < .001), administration of blood products (eg, whole blood, 268 of 2112 [12.7%] vs 0; *P* < .001) and medications (eg, antibiotics, 88 of 2112 [4.2%] vs 58 of 2410 [2.4%]; *P* = .001), were more frequently documented among casualties classified as urgent (eTable 1 in Supplement 1).

Aeromedical evacuation was used significantly more frequently for urgent than nonurgent casualties (1490 of 2112 [70.6%] vs 173 of 2410 [7.2%]; *P* < .001).

Triage Accuracy

The overtriage rate was 62.7% (95% CI, 60.65%-64.77%) (Table 1), reflecting urgent casualties who were either

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Table 1. Characteristics of Patients by Urgency Status[a]

Characteristic	No. (%) Overall (N = 4522)	Urgent (n = 2112)	Nonurgent (n = 2410)	*P* value
Sex				
Female	43 (1.0)	15 (0.7)	28 (1.2)	.12
Male	4479 (99.0)	2097 (99.3)	2382 (98.8)	
Age, median (IQR), y	23 (20-29)	22 (20-28)	23 (20-29)	.003
Casualties per event, No.				
1	1695 (37.5)	533 (25.2)	1162 (48.2)	<.001
2-3	1031 (22.8)	486 (23.0)	545 (22.6)	
≥4	1796 (39.7)	1093 (51.8)	703 (29.2)	
Mechanism of injury[b]				
Explosion	2897 (64.8)	1527 (72.9)	1370 (57.7)	<.001
Firearm	586 (13.1)	413 (19.7)	173 (7.3)	
Blunt/fall	685 (15.3)	87 (4.2)	598 (25.2)	
Other[c]	303 (6.8)	68 (3.2)	235 (9.9)	
No. of injured body regions (field triage)				
1	2898 (64.1)	1001 (47.4)	1897 (78.7)	<.001
2	1004 (22.2)	627 (29.7)	377 (15.6)	
≥3	620 (13.7)	484 (22.9)	136 (5.6)	
Injury localization (field triage)				
Head	519 (11.5)	300 (14.2)	219 (9.1)	<.001
Face	1167 (25.8)	559 (26.5)	608 (25.2)	.30
Neck	378 (8.4)	272 (12.9)	106 (4.4)	<.001
Upper extremities	1710 (37.8)	826 (39.1)	884 (36.7)	.09
Lower extremities	1623 (35.9)	895 (42.4)	728 (30.2)	<.001
Torso	949 (21.0)	672 (31.8)	277 (11.5)	<.001
Pelvis	340 (7.5)	247 (11.7)	93 (3.9)	<.001
Evacuation platform				
Aeromedical evacuation	1663 (36.8)	1490 (70.6)	173 (7.2)	<.001
Severity status (AIS)				
Discharged from ED	2439 (53.9)	474 (22.4)	1965 (81.5)	<.001
Hospitalized, no AIS ≥3	1223 (27.0)	851 (40.3)	372 (15.4)	
Hospitalized, AIS ≥3	860 (19.0)	787 (37.3)	73 (3.0)	
Severity status (ISS)				
Discharged from ED	2439 (53.9)	474 (22.4)	1965 (81.5)	<.001
Hospitalized, ISS <16	1609 (35.6)	1181 (55.9)	428 (17.8)	
Hospitalized, ISS ≥16	474 (10.5)	457 (21.6)	17 (0.7)	
Severity status (NFTI)				
NFTI−	3368 (74.5)	1060 (50.2)	2308 (95.8)	<.001
NFTI+	1154 (25.5)	1052 (49.8)	102 (4.2)	

Abbreviations: AIS, Abbreviated Injury Scale; ED, emergency department; ISS, Injury Severity score; NFTI, need for trauma intervention.

[a] Demographic, injury-related, and clinical characteristics of patients stratified by urgency status at presentation. Injury localization and number of injured body regions reflect findings from field assessment. Injury localization categories are not mutually exclusive; therefore, percentages may sum to more than 100%. Injury severity is presented by anatomic criteria (at least 1 AIS score ≥3), by ISS greater than or equal to 16, and by NFTI.

[b] Missing data for mechanism of injury n = 51 (1.13%).

[c] Other mechanism of injury includes motor vehicle collisions, isolated burn injuries, and superficial low-energy injuries.

discharged from the emergency department or hospitalized without evidence of severe injury (AIS ≥3). The field triage system demonstrated a sensitivity of 91.5% (95% CI, 89.46%-93.19%) for identifying severely injured casualties. When the analysis was restricted to casualties with major trauma (ISS ≥16), sensitivity increased to 96.4% (95% CI, 94.33%-97.75%). Accordingly, undertriage was 8.5% (95% CI, 6.81%-10.54%) among casualties with severe injury and 3.6% (95% CI, 2.25%-5.67%) among those with major trauma. Using an NFTI-based definition, undertriage was 8.8% (95% CI, 7.33%-10.62%), whereas overtriage decreased to 50.2% (95% CI, 48.06%-52.32%).

Among prehospital fatalities (n = 403), 83.6% (95% CI, 79.70%-86.92%) had no signs of life at initial medical contact, while 16.4% (95% CI, 13.08%-20.30%) were evaluated by medical personnel before death and were all classified as urgent. No cases were identified in which a casualty was initially classified as nonurgent and subsequently died during evacuation.

Characteristics of Undertriaged Patients

Among 860 severely injured patients, 73 were undertriaged (**Table 2**). The distribution of ISS categories differed between undertriaged and correctly triaged severely injured patients

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Table 2. Characteristics of Severely Injured Patients by Triage Status[a]

Characteristic	No. (%)			*P* value
	All severely injured [AIS ≥3] (N = 860)	Nonurgent [undertriage] (n = 73)	Urgent (n = 787)	
ISS category				
9-14	386 (44.9)	56 (76.7)	330 (41.9)	<.001
16-24	222 (25.8)	15 (20.5)	207 (26.3)	
≥25	252 (29.3)	2 (2.7)	250 (31.8)	
Mechanism of injury[b]				
Explosion	604 (70.6)	46 (63.9)	558 (71.2)	<.001
Firearm	201 (23.5)	10 (13.9)	191 (24.4)	
Blunt/fall	24 (2.8)	11 (15.3)	13 (1.7)	
Other[c]	27 (3.2)	5 (6.9)	22 (2.8)	
Casualties per event, No.				
1	182 (21.2)	27 (37.0)	155 (19.7)	.002
2-3	175 (20.3)	13 (17.8)	162 (20.6)	
≥4	503 (58.5)	33 (45.2)	470 (59.7)	
Injury localization (AIS ≥3)				
Head	217 (25.2)	22 (30.1)	195 (24.8)	.30
Face	81 (9.4)	0 (0)	81 (10.3)	.004
Neck	22 (2.6)	0 (0)	22 (2.8)	.20
Upper extremities	177 (20.6)	17 (23.3)	160 (20.3)	.50
Lower extremities	312 (36.3)	19 (26.0)	293 (37.2)	.06
Abdomen	141 (16.4)	1 (1.4)	140 (17.8)	<.001
Thorax	277 (32.2)	8 (11.0)	269 (34.2)	<.001
Spine	36 (4.2)	0 (0)	36 (4.6)	.07
External	41 (4.8)	8 (11.0)	33 (4.2)	.02
Evacuation				
Aeromedical evacuation	720 (83.7)	19 (26.0)	701 (89.1)	<.001
Evacuation time, median (IQR), min	67 (51-92)	92 (61-132)	65 (50-87)	<.001
Outcomes				
30-d Mortality	52 (6.0)	0 (0)	52 (6.6)	.02
Hospital length of stay, median (IQR), d[d]	10 (4-23)	3 (2-6)	11 (5-24)	<.001
Discharge home[e,f]	256 (30.1)	53 (72.6)	203 (26.1)	<.001

Abbreviation: AIS, Abbreviated Injury Scale.

[a] Characteristics of patients with severe injury, defined as at least 1 body region with an AIS score of 3 or higher, stratified by triage status at initial assessment. *Undertriage* was defined as patients with severe injury who were initially classified as nonurgent. Injury localization categories include only body regions with AIS score of 3 or higher identified during hospitalization and are not mutually exclusive; therefore, percentages may sum to more than 100%. Percentages were calculated using the total number of patients in each column as the denominator unless otherwise indicated.

[b] Missing data for mechanism of injury 4 (0.47%).

[c] Other mechanism of injury includes motor vehicle collisions, isolated burn injuries, and superficial low-energy injuries.

[d] Missing data for hospital length of stay 15 (1.74%).

[e] Missing data for discharge destination 9 (1.05%).

[f] Discharge home indicates direct discharge from hospitalization to home, excluding transfer to rehabilitation, long-term care facilities or in-hospital death.

(*P* < .001), eg, with 17 of 73 (23.3%) undertriaged casualties having an ISS of 16 or higher compared with 457 of 787 (58.1%) of correctly triaged patients.

Undertriaged casualties were more frequently involved in events with fewer casualties (*P* = .002), eg, with 27 of 73 (37%) occurring in single-casualty events compared with 155 of 787 (19.7%) correctly triaged patients. To further examine this association, we analyzed undertriage according to event size and injury severity (eTable 2 in Supplement 1). Undertriage rates were higher in smaller events (eg, 1 casualty: 27 of 182 [14.8%]; 95% CI, 10.40%-20.72%; *P* = .002) among patients with severe injury (AIS ≥3), however, not among those with major trauma (ISS ≥16), for whom undertriage rates did not vary by event size (eg, 1 casualty: 4 of 87 [4.6%]; 95% CI, 1.80%-11.23%; *P* = .73).

The distribution of injury mechanisms also differed between severely injured casualties designated as nonurgent and those designated as urgent (*P* < .001), eg, with blunt trauma or falls occurring in 11 of 72 (15.3%) undertriaged casualties vs 13 of 784 (1.7%) correctly triaged casualties.

Hospital data showed lower rates of severe thoracic (8 of 73 [11.0%] vs 269 of 787 [34.2%]; *P* < .001), abdominal (1 of 73 [1.4%] vs 140 of 787 [17.8%]; *P* < .001), and facial (0 vs 81 of 787 [10.3%]; *P* = .004) injuries among undertriaged patients. In contrast, external injuries with an AIS of 3 or higher, which include severe soft-tissue injuries irrespective of anatomical location, were more frequent among undertriaged patients than among correctly triaged casualties with severe injuries (8 of 73 [11.0%] vs 33 of 787 [4.2%]; *P* = .02).

These patients were less likely to be evacuated by air (19 of 73 [26.0%] vs 701 of 787 [89.1%]; *P* < .001) and experienced significantly longer median (IQR) evacuation times of 92 (61-132) minutes vs 65 (50-87) minutes (*P* < .001). Despite these differences, no 30-day mortality was observed in the undertriaged group, as compared with 52 of 787 (6.6%) in the correctly triaged urgent group (*P* = .02). Furthermore, among undertriaged casualties, 53 of 73 (72.6%) were discharged home (ie, discharged directly home without transfer to rehabilitation or long-term care facilities), compared with 203 of 778 (26.1%) of those patients

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Table 3. Manual Review of Undertriaged Casualties with Major Trauma (ISS ≥16)[a]

Case No.	Age, y	Total casualties per event, No.	Mechanism of injury	Injuries identified in field	GCS score	Severe injury
1	21	3	Explosion	Blast abdominal injury; blunt head trauma; acoustic trauma	15	Intracranial hemorrhage
2	33	1	Explosion	Blunt chest and pelvic trauma	15	Rib fractures
3	19	4	Explosion	Blast ear and neck injuries; blunt hand injury	15	Intracranial hemorrhage; skull fractures
4	20	8	Explosion	Finger amputation; neck shrapnel injury	15	Lung contusion
5	21	7	Explosion	Facial and thoracic shrapnel injury	15	Intracranial hemorrhage
6	21	1	Explosion	Facial shrapnel injury	15	Intracranial hemorrhage
7	20	1	Explosion	Shrapnel head injury	15	Intracranial hemorrhage; skull fractures
8	39	7	Explosion	Lower limb shrapnel injury; animal bite	15	Severe lower limb vascular injury
9	36	5	Explosion	Multiorgan shrapnel injuries (head, neck, thorax, extremities, pelvis)	15	Penetrating head injury; lung contusion
10	29	5	Explosion	Extensive burns; acoustic injury	NA[b]	Burn severity; inhalation injury
11	21	12	Explosion	Facial shrapnel injury	NA[b]	Lung contusion
12	21	4	Explosion	Extensive burns	15	Burn severity
13	21	2	Explosion	Scalp and arm shrapnel injury	15	Intracranial hemorrhage; skull fractures
14	33	3	Firearm	Arm gunshot wound	15	Limb fracture with nerve injury; pneumomediastinum
15	23	11	Explosion	Head shrapnel injury; acoustic injury	15	Penetrating head injury
16	21	1	Explosion	Facial shrapnel injury	15	Intracranial hemorrhage; skull fractures
17	25	2	Explosion	Blunt head injury	15	Intracranial hemorrhage; skull fractures

Abbreviations: AIS, Abbreviated Injury Scale; GCS, Glasgow Coma scale; ISS, Injury Severity score; NA, numeric score not documented.

[a] Detailed clinical characteristics of patients with major trauma (ISS ≥16) who were initially classified as nonurgent (undertriage). Each row represents an individual patient. Injuries identified in the field reflect the initial clinical assessment at the scene, whereas severe injury refers to injuries meeting criteria for severe trauma (AIS score ≥3). GCS scores were recorded at initial assessment.

[b] Patient was described as alert.

correctly triaged and severely injured ($P < .001$). Their median (IQR) hospital length of stay was 3 (2-6) days vs 11 (5-24) days in correctly triaged patients ($P < .001$).

To further investigate the nature of undertriage among patients with major trauma, an in-depth review was conducted for the 17 casualties (3.6%) with ISS of 16 or higher (**Table 3**). Head trauma emerged as the most frequently overlooked injury, with severe injuries identified in 10 patients (59%), including 8 cases of intracranial hemorrhage and 5 with associated skull fractures. Additional severe injuries in this population included thoracic trauma (5 [29%]) and, less frequently, burns (2 [12%]) and injuries to the upper (1 [6%]) or lower (1 [6%]) extremities (**Figure 2**).

Undertriage rates did not change significantly over time during the study period for either severe or major trauma (eFigure in Supplement 1).

Evacuation Prioritization Among Urgent Casualties

To evaluate the operational impact of overtriage, we examined the association between injury severity and evacuation time among hospitalized urgent casualties. The most severely injured patients had the shortest evacuation times, with progressively longer times observed across decreasing injury severity (from median [IQR] of 62 [48-79] minutes for ISS of 25 or higher to 78 [61-109] minutes for ISS of 1-8), showing a graded inverse relationship across ISS categories ($P < .001$) (eTable 3 in Supplement 1). This pattern persisted in events involving 4 or more casualties.

Discussion

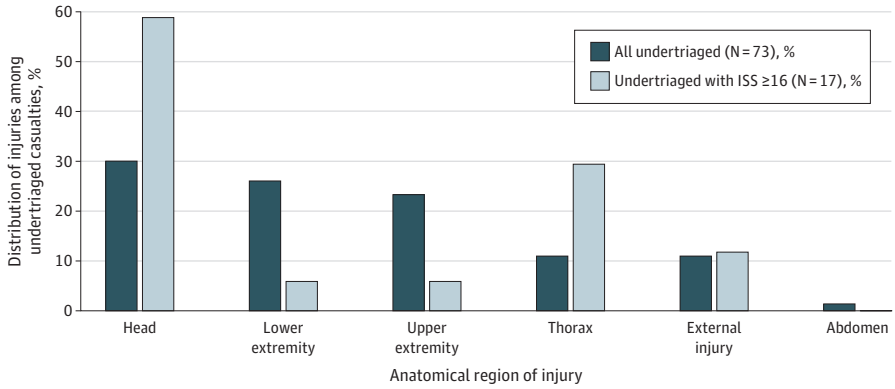
Results of this large-scale retrospective analysis of 4522 military trauma casualties suggest that a simplified binary field triage system can achieve high sensitivity for identifying severe and major trauma in combat settings. Using an anatomic definition of severe injury (AIS ≥3), the system demonstrated a sensitivity of 91.5% increasing to 96.4% among patients with major trauma (ISS ≥16), corresponding to undertriage rates of 8.5% and 3.6%, respectively. These findings indicate that a simplified, judgment-based approach can effectively identify injured patients in operational environments, albeit with substantial overtriage (62.7%), reflecting a deliberate prioritization of sensitivity. Similar results were observed using an intervention-based definition (NFTI), with undertriage and overtriage rates of 8.8% and 50.2%, respectively, supporting the robustness of this approach.

Commonly cited field triage benchmarks include an undertriage rate of 5% and overtriage rates of 25% to 50%.[4,19,20] In the present study, overtriage exceeded these ranges, whereas undertriage for major trauma approached the recommended threshold, consistent with prior high-sensitivity strategies that report similarly elevated overtriage.[19] Although such a tradeoff may be acceptable in settings with high resource availability and short evacuation times, as in the present conflict, it may impose a substantial burden on evacuation systems in large-scale combat operations, where evacuation may be

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Figure 2. Bar Graph and Chart Showing the Distribution of Injury Locations Among Undertriaged Patients



Anatomical region head	All undertriaged (N = 73), %	All undertriaged, No.	Undertriaged with ISS ≥16 (N = 17), %	Undertriaged with ISS ≥16, No.
Head	30	22	59	10
Lower extremity	26	19	6	1
Upper extremity	23	17	6	1
Thorax	11	8	29	5
External injury	11	8	12	2
Abdomen	1	1	0	0

Proportion of patients with severe injury (Abbreviated Injury Scale score ≥3) by anatomical region among undertriaged casualties. ISS indicates Injury Severity Score.

constrained.[21] Under such conditions, high overtriage may misallocate limited resources and delay care for critically injured patients.

Undertriaged casualties were generally less severely injured. Most had moderate injury severity (ISS of 9-14), typically reflecting isolated injuries, predominantly orthopedic trauma such as extremity fractures. Less overt mechanisms, particularly blunt trauma, were relatively more frequent, consistent with prior reports.[22,23] Undertriage was also more common in single-casualty events. However, among patients with major trauma, rates were similar across event sizes, suggesting this may be associated with differences in injury severity distribution. Accordingly, outcomes were more favorable compared with correctly triaged patients, with no 30-day mortality and most discharged home, indicating limited short-term clinical impact. These findings suggest that the binary nature of the system may limit classification of patients with moderate or less apparent injury severity.

At the same time, undertriage was not clinically trivial. Seventeen patients met criteria for major trauma, and in this subgroup, head injuries were most frequently overlooked (59%), including intracranial hemorrhage and skull fractures, consistent with prior literature.[24,25] Additional missed injuries included thoracic trauma, underscoring the difficulty of identifying certain high-risk patterns, particularly following blast exposure.[26] These findings suggest that suspected head injury may warrant increased clinical attention.

Overtriage also has important operational implications. Approximately 22% of casualties classified as urgent were ultimately discharged from the ED, reflecting the trade-off between sensitivity and specificity.[27] However, among hospitalized urgent casualties, a graded inverse association was observed between injury severity and evacuation time. This pattern persisted even in higher-casualty events, suggesting that evacuation prioritization within the urgent category may partially mitigate the adverse effects of overtriage. This may reflect the proximity of well-trained ALS practitioners to the point of injury, enabling early and dynamic assessment and prioritization.[14]

Comparison with prior literature should be made cautiously, as most studies derive from civilian trauma systems where undertriage remains a persistent challenge. In a large prospective study, undertriage was 21.6%, increasing to 63.8% under strict protocol adherence.[28] A multivariable model improved sensitivity to 88.8%, but undertriage remained near 10% and required greater operational complexity.[10] In combat settings, complex models may be less feasible, as they typically involve limited data, time pressure, and ongoing threat, often relying on a single early assessment.[6] Under such conditions, a binary approach may provide a pragmatic balance between feasibility and accuracy.

Limitations

This study has several limitations. First, it focused exclusively on military trauma casualties in a combat setting, which may limit generalizability to other trauma populations. In addition, the study reflects a specific conflict with relatively high resource availability. The performance of this triage system under conditions of extreme casualty load, such as large-scale combat operations, was not evaluated and may differ.

Second, estimates of undertriage and overtriage are inherently constrained by the structure of the registry, as casualties discharged from the ED without hospitalization are not captured in the INTR and were, therefore, classified as nonseverely injured. This approach may have led to

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misclassification of patients whose injury severity was underestimated. Furthermore, although no 30-day mortality was observed among undertriaged patients, the absence of long-term functional outcome data in the registry limits direct assessment of morbidity associated with undertriage. In addition, prehospital fatalities were excluded because they are not captured in the registry. However, they were analyzed separately with respect to field triage designation.

Despite these limitations, the analyzed cohort represents the majority of trauma patients in the current conflict and provides a robust basis for evaluating field triage performance.

Conclusions

In this cohort study, the simplified IDF-MC triage system demonstrated high sensitivity for identifying severe injury in combat settings using clinical judgment without complex scoring. This approach prioritizes minimizing missed severe injury at the cost of increased overtriage. Its performance in large-scale, resource-constrained settings requires further evaluation. Future efforts should focus on hybrid approaches that preserve simplicity while incorporating technologies[29,30] to enhance accuracy.

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REFERENCES

1. Bhaumik S, Hannun M, Dymond C, et al. Prehospital triage tools across the world: a scoping review of the published literature. *Scand J Trauma Resusc Emerg Med*. 2022;30(1):32. doi:10.1186/s13049-022-01019-z

2. Sharma BR. Development of prehospital trauma-care system—an overview. *Injury*. 2005;36(5):579-587. doi:10.1016/j.injury.2004.09.009

3. Newgard CD, Fischer PE, Gestring M, et al; Writing Group for the 2021 National Expert Panel on Field Triage. National guideline for the field triage of injured patients: Recommendations of the National Expert Panel on Field Triage, 2021. *J Trauma Acute Care Surg*. 2022;93(2):e49-e60. doi:10.1097/TA.0000000000003627

4. van Rein EAJ, Houwert RM, Gunning AC, Lichtveld RA, Leenen LPH, van Heijl M. Accuracy of prehospital triage protocols in selecting severely injured patients: a systematic review. *J Trauma Acute Care Surg*. 2017;83(2):328-339. doi:10.1097/TA.0000000000001516

5. Morris RS, Karam BS, Murphy PB, et al. Field-triage, hospital-triage and triage-assessment: a literature review of the current phases of adult trauma triage. *J Trauma Acute Care Surg*. 2021;90(6):e138-e145. doi:10.1097/TA.0000000000003125

6. Falzone E, Pasquier P, Hoffmann C, et al. Triage in military settings. *Anaesth Crit Care Pain Med*. 2017;36(1):43-51.

7. Benov A, Glassberg E, Nadler R, et al. Role I trauma experience of the Israeli Defense Forces on the Syrian border. *J Trauma Acute Care Surg*. 2014;77(3)(suppl 2):S71-S76. doi:10.1097/TA.0000000000000377

8. Peng J, Xiang H. Trauma undertriage and overtriage rates: are we using the wrong formulas? *Am J Emerg Med*. 2016;34(11):2191-2192. doi:10.1016/j.ajem.2016.08.061

9. Baker SP, O'Neill B, Haddon W Jr, Long WB. The injury severity score: a method for describing patients with multiple injuries and evaluating emergency care. *J Trauma*. 1974;14(3):187-196. doi:10.1097/00005373-197403000-00001

10. van Rein EAJ, van der Sluijs R, Voskens FJ, et al. Development and validation of a prediction model for prehospital triage of trauma patients. *JAMA Surg*. 2019;154(5):421-429. doi:10.1001/jamasurg.2018.4752

11. Morris RS, Davis NJ, Koestner A, Napolitano LM, Hemmila MR, Tignanelli CJ. Redefining the trauma triage matrix: the role of emergent interventions. *J Surg Res*. 2020;251:195-201. doi:10.1016/j.jss.2019.11.011

12. Roden-Foreman JW, Rapier NR, Yelverton L, Foreman ML. Asking a better question: development and evaluation of the need for trauma intervention (NFTI) metric as a novel indicator of major trauma. *J Trauma Nurs*. 2017;24(3):150-157. doi:10.1097/JTN.0000000000000283

13. Tsur AM, Nadler R, Lipsky AM, et al. The Israel Defense Forces Trauma Registry: 22 years of point-of-injury data. *J Trauma Acute Care Surg*. 2020;89(2S)(suppl 2):S32-S38. doi:10.1097/TA.0000000000002776

14. Benov A, Glassberg E, Baruch EN, et al. Augmentation of point of injury care: reducing battlefield mortality—the IDF experience. *Injury*. 2016;47(5):993-1000. doi:10.1016/j.injury.2015.10.078

15. Aharonson-Daniel L, Avitzour M, Giveon A, Peleg K. A decade to the Israel National Trauma Registry. *Isr Med Assoc J*. 2007;9(5):347-351.

16. Regev S, Talmy T, Gelikas S, Mitchnik IY. Impact of ongoing clinical experience on advanced life support provider performance in multicasualty simulations. *J Trauma Acute Care Surg*. 2025;99(3s suppl 1):s32-s38. doi:10.1097/TA.0000000000004697

17. Uleberg O, Vinjevoll OP, Eriksson U, Aadahl P, Skogvoll E. Overtriage in trauma—what are the causes? *Acta Anaesthesiol Scand*. 2007;51(9):1178-1183. doi:10.1111/j.1399-6576.2007.01414.x

18. Mohan D, Barnato AE, Rosengart MR, et al. Trauma triage in the emergency departments of

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nontrauma centers: an analysis of individual physician caseload on triage patterns. *J Trauma Acute Care Surg*. 2013;74(6):1541-1547. doi:10.1097/TA.0b013e31828c3f75

19. Lupton JR, Davis-O'Reilly C, Jungbauer RM, et al. Undertriage and overtriage using the field triage guidelines for injured patients: a systematic review. *Prehosp Emerg Care*. 2023;27(1):38-45. doi:10.1080/10903127.2022.2043963

20. Newgard CD, Fu R, Lerner EB, et al. Role of guideline adherence in improving field triage. *Prehosp Emerg Care*. 2017;21(5):545-555. doi:10.1080/10903127.2017.1308612

21. Remondelli MH, Remick KN, Shackelford SA, et al. Casualty care implications of large-scale combat operations. *J Trauma Acute Care Surg*. 2023;95(2s suppl 1):s180-s184. doi:10.1097/TA.0000000000004063

22. Anderson K, Schellenberg M, Owattanapanich N, et al. Undertriage of severely injured trauma patients. *Am Surg*. 2023;89(10):4129-4134. doi:10.1177/00031348231177939

23. Johnston TJ, Filiberto DM, DePhillips PB, et al. Assessing the performance of the updated 2021 Field Triage Guidelines with the need for trauma intervention (NFTI) metric. *Am J Surg*. 2026;252:116729. doi:10.1016/j.amjsurg.2025.116729

24. Nakahara S, Matsuoka T, Ueno M, et al. Predictive factors for undertriage among severe blunt trauma patients: what enables them to slip through an established trauma triage protocol? *J Trauma*. 2010;68(5):1044-1051. doi:10.1097/TA.0b013e3181aca144

25. Xiang H, Wheeler KK, Groner JI, Shi J, Haley KJ. Undertriage of major trauma patients in the US emergency departments. *Am J Emerg Med*. 2014;32(9):997-1004. doi:10.1016/j.ajem.2014.05.038

26. Bukowski J, Nowadly CD, Schauer SG, Koyfman A, Long B. High risk and low prevalence diseases: Blast injuries. *Am J Emerg Med*. 2023;70:46-56. doi:10.1016/j.ajem.2023.05.003

27. Fuller GW, Baird J, Keating S, et al; and the MATTS research group. Major Trauma Triage Study (MATTS): Diagnostic accuracy of major trauma triage tools in English regional trauma networks: a case-cohort study. *PLoS One*. 2026;21(3):e0344996. doi:10.1371/journal.pone.0344996

28. Voskens FJ, van Rein EAJ, van der Sluijs R, et al. Accuracy of prehospital triage in selecting severely injured trauma patients. *JAMA Surg*. 2018;153(4):322-327. doi:10.1001/jamasurg.2017.4472

29. Lokerman RD, van der Sluijs R, Waalwijk JF, et al; Prehospital Trauma Triage Research Collaborative (PTTRC). Development and validation of prediction models for prehospital triage of military trauma patients. *BMJ Mil Health*. 2025;171(e1):e56-e62. doi:10.1136/military-2023-002644

30. Frock A, Robbins JT, Vital-Lopez FG, et al. A case study of AI-enabled software as a medical device cleared by the FDA for assessing hemorrhage risk index (APPRAISE-HRI) after trauma. *NEJM AI*. 2025;2(11). doi:10.1056/AIcs2401170

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